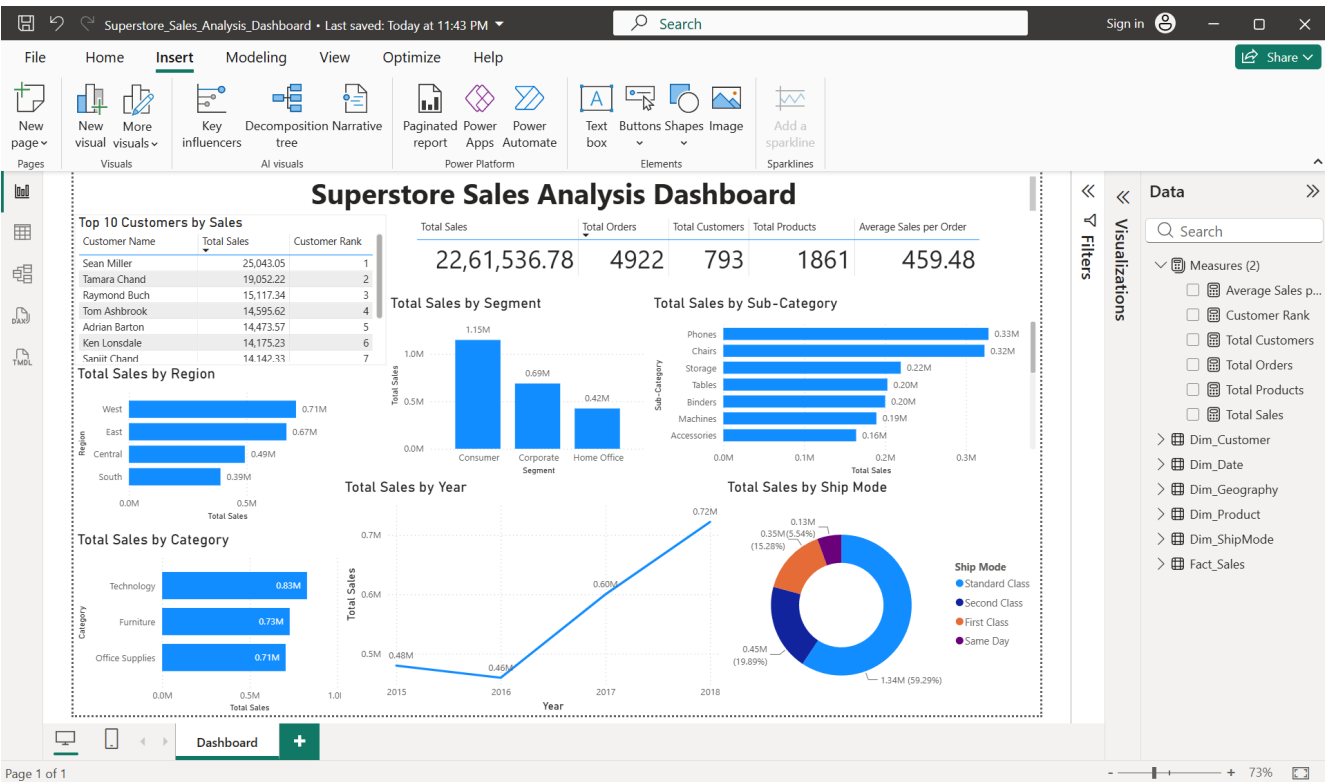


Superstore Sales Analysis Report

Excel | SQL | Power BI | Power Query | DAX

Prepared by Shubham Kumar Dubey



Dashboard Preview

1. Executive Summary

This report presents the final business analysis of the Superstore sales dataset. The project was completed using Excel for initial validation, SQL for analytical queries, and Power BI for data modeling, DAX measures, and dashboard reporting. The analysis focuses on sales performance by region, category, customer segment, product sub-category, ship mode, yearly trend, and top customers.

2. Key Performance Indicators

KPI	Value
Total Sales	2,261,536.78
Total Orders	4,922
Total Customers	793
Total Products	1,861
Average Sales per Order	459.48

3. Methodology

- Cleaned and validated the dataset in Excel.
- Performed SQL analysis in Google Colab using SQLite to verify totals, rankings, and business trends.
- Built a Power BI star schema model using one fact table and multiple dimension tables.
- Created DAX measures for sales, orders, customers, products, average sales per order, and customer ranking.
- Designed a Power BI dashboard to summarize business performance clearly.

4. Dataset and Data Model

The dataset contains 9,800 sales records and 18 original columns, including order details, customer information, product details, geography fields, shipping mode, and sales amount.

The Power BI model was structured using a star schema for better analysis and cleaner reporting.

Table	Description
Fact_Sales	Transaction-level sales data with Order ID, dates, keys, geography key, and Sales.
Dim_Customer	Customer ID, Customer Name, and Segment.
Dim_Product	Product ID, Product Name, Category, and Sub-Category.
Dim_Geography	Country, City, State, Postal Code, Region, and GeoKey.
Dim_ShipMode	Shipping mode information.
Dim_Date	Date, Year, Month, and Month Name.

5. Power BI Dashboard

The dashboard includes KPI summary, regional sales, category sales, segment sales, yearly sales trend, sub-category sales, ship mode distribution, and top customer ranking.

6. Key Business Insights

6.1 Sales by Region

Metric	Total Sales
West	710,219.68
East	669,518.73
Central	492,646.91
South	389,151.46

Insight: West is the highest-performing region, while South has the lowest sales among all regions.

6.2 Sales by Category

Metric	Total Sales
Technology	827,455.87
Furniture	728,658.58
Office Supplies	705,422.33

Insight: Technology is the top-performing category and contributes the highest revenue.

6.3 Sales by Customer Segment

Metric	Total Sales
Consumer	1,148,060.53
Corporate	688,494.07
Home Office	424,982.18

Insight: Consumer customers generate the highest sales, making them the strongest customer segment.

6.4 Sales by Ship Mode

Metric	Total Sales
Standard Class	1,340,831.31
Second Class	449,914.18
First Class	345,572.26
Same Day	125,219.04

Insight: Standard Class is the dominant shipping mode and contributes the highest sales.

6.5 Top Sub-Categories

Metric	Total Sales
Phones	327,782.45
Chairs	322,822.73
Storage	219,343.39
Tables	202,810.63
Binders	200,028.79

Insight: Phones and Chairs are the strongest sub-categories by total sales.

6.6 Top Customers

Rank	Customer Name	Total Sales
1	Sean Miller	25,043.05
2	Tamara Chand	19,052.22
3	Raymond Buch	15,117.34
4	Tom Ashbrook	14,595.62
5	Adrian Barton	14,473.57

Insight: Sean Miller is the highest-value customer. However, his contribution is only about 1.11% of total sales, meaning revenue is distributed across many customers instead of depending on one customer.

7. SQL Analysis Summary

SQL was used to validate and analyze the dataset before creating the Power BI dashboard. The SQL analysis covered total sales, total orders, total customers, sales by region, category, segment, sub-category, monthly sales trend, customer ranking, regional top customers, customer revenue contribution percentage, top products, and ship mode performance.

The detailed SQL queries are kept separately in the SQL notebook file so that this report remains focused on business insights and final dashboard interpretation.

8. Final Conclusion

The Superstore analysis shows total sales of 2.26M from 4,922 orders and 793 customers. West is the strongest region, Technology is the highest-performing category, and the Consumer segment generates the highest revenue. Phones and Chairs are the top sub-categories, while Standard Class is the leading ship mode. The customer base is distributed, with the top customer contributing only 1.11% of total sales.

9. Skills Demonstrated

- Excel data cleaning and validation
- SQL aggregation and window functions
- Power Query transformation
- Star schema data modeling
- DAX measure creation
- Power BI dashboard design
- Business insight generation
- GitHub project documentation

10. Author

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Aspiring Data Analyst skilled in Excel, SQL, Power BI, Power Query, DAX, and dashboard reporting.